

A commutative diagram illustrating a relationship between four mathematical objects arranged in a square:

- Top-left node: U
- Top-right node: $f(U)$
- Bottom-left node: $\mathbb{C}$
- Bottom-right node: $\mathbb{C}$

The maps connecting these nodes are:

- Horizontal map from U to $f(U)$: f
- Horizontal map from $\mathbb{C}$ to $\mathbb{C}$: $z \mapsto z^n$
- Vertical map from U to $\mathbb{C}$: ϕ
- Vertical map from $f(U)$ to $\mathbb{C}$: ϕ

The diagram is a square with arrows pointing from top-left to top-right, top-right to bottom-right, bottom-right to bottom-left, and bottom-left to top-left, forming a cycle.